Ranking the Greatest F1 Drivers: How Good Were They Without the Car?

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Summary

Once the car is taken out, no single driver stands clear at his best. Five share the top tier at peak: Fangio, Senna, Clark, Michael Schumacher and Jackie Stewart. Over a whole career from 1970, Schumacher is the strongest, with a 70% chance of ranking first on the 50/50 measure.

The model was built from every Formula 1 race and qualifying session from 1950 to 2026. It treats each result as a contest between drivers who were each given a car of a certain speed. It estimates how fast each team's car was every season and how good each driver was in any season. It then puts every driver in identical cars and simulates thousands of race weekends. A driver's score is his average finishing position in that simulation, in a 20-car field, where lower is better.

Measure	Tier 1	Tier 2
Peak (best 3 straight seasons), all-time from 1950	Fangio, Senna, Clark, Schumacher, Stewart	G. Hill, Verstappen, Lauda, Rindt, Alonso, Mansell, Piquet, Ascari
Combined (50% peak, 50% career), from 1970	Schumacher	Alonso, Verstappen, Senna, Hamilton
Career (total value over a career), from 1970	Alonso, Schumacher, Verstappen, Hamilton	Senna

Four findings matter most:

- **Drivers can be ranked by tier, not by exact place.** Inside a tier the order is close to a coin toss, so the honest answer is a tier.
- **The model backs the case for drivers without the best car.** In 2012, Alonso gained more over an average driver in his seat than anyone else, in only the third-best car.
- **The earliest drivers need a warning.** Before the late 1960s the model sometimes rates a car too low and gives the extra credit to its driver. Fangio and Graham Hill are the clearest cases.
- **The best single season is Schumacher's 1994,** narrowly ahead of Fangio's 1953 and 1954.

1. The question

The aim is to rank Formula 1 drivers by how well they drove, relative to their car and their teammates, not by what they won.

Championships mostly measure the car. In most seasons the title goes to a driver in one of the two fastest cars, so counting titles largely ranks a driver's luck in picking teams. Fernando Alonso in 2012 shows the problem. He took a Ferrari that was only the third-fastest car to the final race of the championship, which arguably shows more skill than winning in the best car. A fair ranking has to credit that.

Two questions run through the whole project:

1. **How good was a driver at his best?** This is his *peak*: his best three consecutive seasons.
2. **How much value did he add across his whole career?** This is his *career* score, which rewards being very good for a long time.

They give different answers, so they are kept separate, with a 50/50 combined ranking alongside. A brilliant short career, like Jim Clark's or Ayrton Senna's, scores highly on peak. A long, consistently strong one, like Alonso's or Lewis Hamilton's, scores highly on career.

2. The data

Every qualifying session and race from 1950 to 2026 comes from F1DB, an open database of official results. Lap times from 2018 come from TracingInsights. Four lists were added by hand, because no database records them.

Source	What it gives	Used for
F1DB results, 1950–2026	Qualifying and finishing positions, retirements and their causes, teams, sprints	The main model
TracingInsights laps, 2018–2026	Every lap time and the tyre on each lap	A separate lap-time model, used only as an independent check
Wet-race list	Share of each race run on intermediate or wet tyres	Separating wet and changeable skill from dry skill
Team-orders list	16 known cases where a team swapped or held its drivers	Correcting results that weren't real contests
Collision flags	Retirements caused by contact	Kept out of the scores and reported as uncertainty

Source	What it gives	Used for
Car lineage	Customer teams that bought another team's chassis	Linking those cars to the right team

How the wet list was built. A race counts as *wet* if at least half of it was run on intermediate or wet tyres, and as *changeable* if 20–50% was. Below that it is dry. From 2018 the tyre data gives the exact share: 9 wet races and 7 changeable ones. Before 2018 there is no tyre data, so I estimated 116 races from 1971 to 2017 from race reports, giving 44 wet and 35 changeable. On six 2018+ races where both were available, the estimate was within 7 percentage points of the truth. The seven 1966–68 candidates were checked by hand, and Britain 1966 turned out to be dry.

What was cleaned. Results where two drivers shared one car in the 1950s are left unranked, because they don't say who drove better. The Indianapolis 500 is excluded from 1950–60: it counted for the championship, but almost no regular F1 drivers entered it. Sprints count as separate, shorter results, and the model was also re-run without them.

3. How the model works

The model explains each result as a contest between drivers. Each driver brings two things: his own ability and the speed of his car. Everything else is treated as luck.

Teammates are the key comparison

Two teammates drive the same car, so the gap between them is down to the drivers. Drivers also change teams, so these comparisons link up into one chain. Alonso was measured against Massa at Ferrari, Massa against Räikkönen, Räikkönen against Hamilton's teammates at McLaren, and so on back through the decades. Once every driver is placed on that chain, the remaining differences between teams are the cars.

The race model (Plackett–Luce)

Each driver in a session gets a *performance score*: his ability plus his car's speed. The result is then treated as a draw. The winner is picked first, with drivers on higher scores more likely to be picked. Then second place is picked from those left, and so on down the field. This is a Plackett–Luce model, widely used for ranked results. For a finishing order of drivers 1, 2, …, n with performance scores u:

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P(\text{order}) = \prod_{k=1}^{n} \frac{e^{u_k}}{\sum_{j \ge k} e^{u_j}}
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The gap between two scores sets how often one driver beats the other. A gap of 1 means the better driver wins a straight head-to-head about 73% of the time.

What goes into a performance score

Part	What it captures	How it is modelled
Driver skill	How good the driver is, separately in qualifying and in races	One value per driver
Driver form	Changes from season to season, for example improvement, decline or a comeback	Each season starts from the previous one. A gap of two or more years allows a bigger change, for example Schumacher's return in 2010
Car speed	How fast each team's car was that season	One value per team-season. Each car carries on from the team's previous car (about 0.8 of last year's edge), so a dominant team stays strong without one-off seasons being flattened
Wet and changeable skill	Extra pace, or lack of it, when it rains	A driver-specific add-on that switches on in wet or changeable sessions
Adapting	A temporary dip after changing team or after a major rules change	A driver-specific penalty that fades over about 6 races, measured against the typical driver
Grid position	Starting at the front helps in the race	One effect per era, since overtaking has been easier or harder at different times
Driver errors	Crashing or spinning out	A separate retirement rate for each driver. Mechanical failures are not held against the driver

Qualifying and race results are modelled together. Race skill is tied to qualifying skill but can differ from it, so a driver can be faster over one lap than over a race distance or the other way round.

Fitting it: Bayesian estimation

The model was fitted with Bayesian methods, which give a full range of plausible values for every driver and car rather than one number. In practice it ran thousands of steps of a sampling algorithm (NUTS) in two independent chains. If both chains end up describing the same answer, the fit has settled. That is checked with a statistic called R-hat, where 1.00 is perfect and below about 1.05 is acceptable. The final models reached 1.001–1.013 on their main values.

The ranges matter as much as the averages. Every ranking below comes with a range of places a driver could plausibly hold, and the tiers are built from those ranges.

Four versions of the model

Version	Data	Role
C50	Results, 1950–2026	All-time peak ranking
B80	Results, fitted from 1966, reported from 1970	Career and combined rankings
B18	Results, 2018–2026	Like-for-like comparison with the lap-time model
A18	Lap times, 2018–2026	Independent check that sees pace, not just positions

Career and combined rankings start in 1970 because the earliest seasons were short and patchy. A driver racing from 1950 to 1957 had far fewer races to build a career total from, and the data on him is thinner. Peak is a rate rather than a total, so it can be compared fairly from 1950.

4. Turning the model into scores

The raw model values are on an abstract scale, so they are converted into something readable: a finishing position in identical cars.

The identical-car simulation

Every qualifying driver-season from 1950 to 2026 goes into one shared field. Fangio's 1954, Senna's 1988 and Verstappen's 2023 all line up together. The cars are made identical, and each driver keeps his own qualifying and race skill, wet skill and error rate. Each simulated

weekend has a qualifying session, then a race started from that grid. Some weekends are wet, at the real historical rate. The simulation runs 24 weekends for each of 120 plausible versions of the model, and each driver's finishes are averaged.

The result is rescaled to a 20-car field. **A season score of 5.0 means the driver would average fifth place against everyone in F1 history in equal cars.** Lower is better, and the average driver scores about 10.5.

Because every era races in one field, a weak field of amateurs in the 1950s doesn't make Fangio's score look better. He is measured against the whole history of the sport, not just his contemporaries.

A season only counts if the driver started at least 8 races, or 60% of a short early calendar.

Peak

Peak is a driver's best average over three consecutive seasons. Three seasons smooths out one lucky year but still captures a driver at his best. Longer careers get slightly more chances to find a good window. Each extra season flatters a driver by about 0.01–0.03, which was measured and judged small.

Career

Career adds up how far each season was above an average driver, weighted by the races started:

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\text{Career} = \sum_{\text{seasons}} (10.5 - \text{season score}) \times \frac{\text{races}}{22}
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A driver who averaged 3 places better than average for 10 full seasons scores about +30. Mediocre seasons add little, and below-average seasons subtract.

Combined

The combined ranking gives peak and career equal weight. Each is scaled so the best driver on it scores 1, then the two are averaged. Rankings with 75% and 25% on peak were also produced for comparison.

Tiers

The ranges show that exact places can't be trusted, so drivers are grouped into tiers. The leader of the first tier is the driver with the best typical rank. Anyone with at least a 25% chance of really being better than that leader joins the tier. The next tier starts from the

best driver left over, and so on. Tiers are kept in rank order, so no driver jumps a tier above someone ranked better.

Left out on purpose: driver-team fit

The model could estimate how well each driver suited each team. That was left out of the scores, because a long stay at a strong team would soak up car speed and get credited as fit.

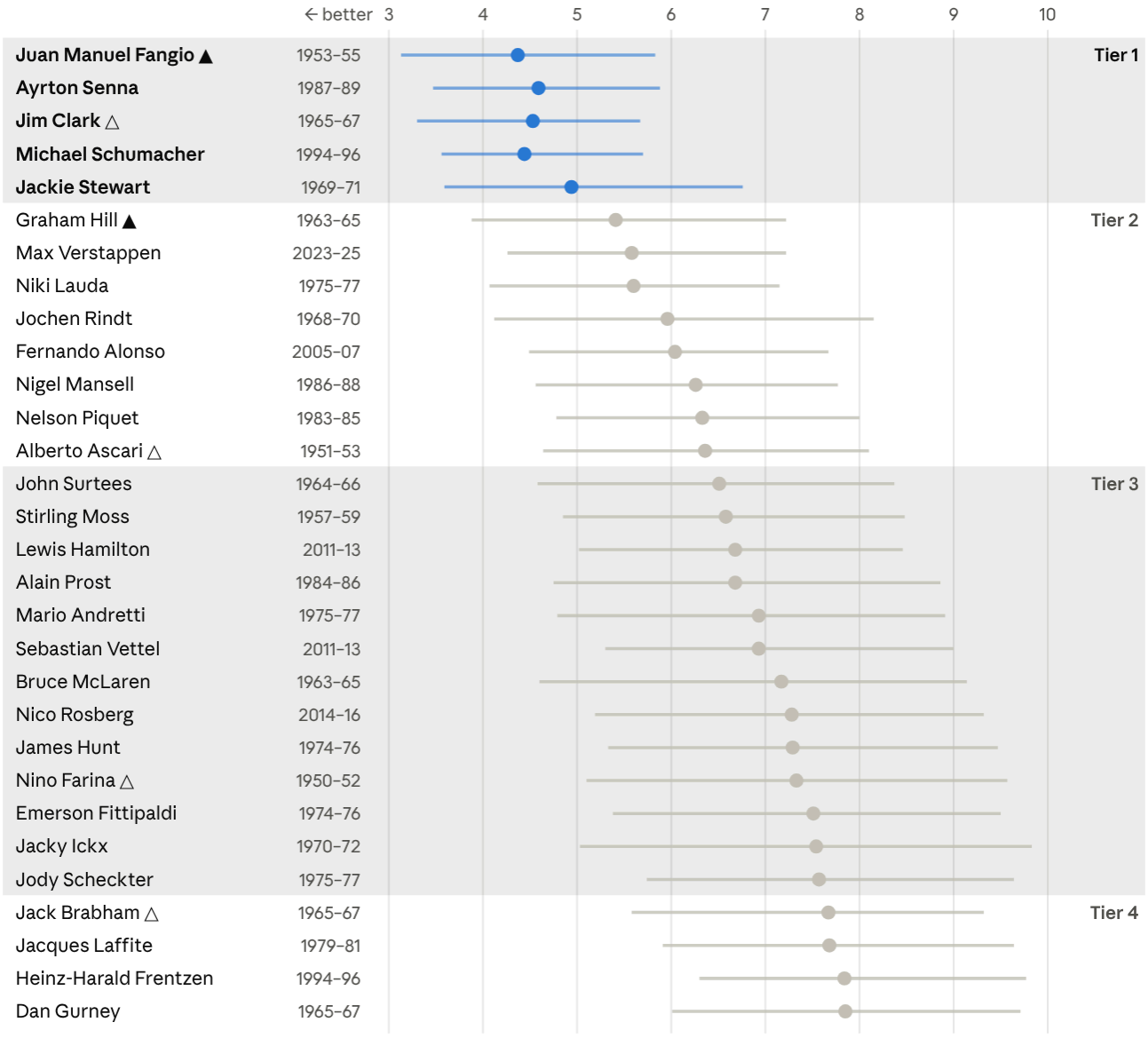
5. The final ranking

The top of every ranking is a group, not a single name. The rankings below are final. For each driver, *typical rank* is the middle of his plausible places, and the *90% range* is the span his rank falls in nine times out of ten.

All-time peak, from 1950

Five drivers share the top tier at peak, and their ranges overlap almost completely

Peak score: average finish in identical cars over the best three straight seasons, 1950–2026.
Dot = best estimate, line = 90% range. Lower is better.



▲ likely inflated, △ possibly inflated, by a car the model rates too low (section 8)

C50 model (1950–2026), drivers with 25+ starts · top 30 by typical rank

The five in tier 1 each have only a 10–24% chance of being the single best: Fangio 24%, Schumacher 16%, Clark 15%, Senna 13% and Stewart 10%. Their best estimates sit within 0.6 of a place of each other. Verstappen leads the current era and is at the front of tier 2. Hamilton is in tier 3 at peak, partly because Rosberg ran him close in the 2014–16 Mercedes. That leads the model to credit much of that era's speed to the car. Drivers who had no close teammate in their best car didn't face that test, so they look better (section 8).

Combined: 50% peak and 50% career, from 1970

Tier	Driver	Peak (best 3 seasons)	Career	Typical rank	90% range
1	**Michael Schumacher**	4.48 (1994–96)	+56.5	1	1–3
2	**Max Verstappen**	5.27 (2024–26)	+48.4	3	1–5
2	Fernando Alonso	6.02 (2006–08)	+52.3	3	1–7
2	Ayrton Senna	4.75 (1987–89)	+36.5	4	2–6
2	Lewis Hamilton	6.61 (2010–12)	+44.3	5	2–11
3	**Niki Lauda**	5.68 (1976–78)	+26.7	6	3–13
3	Nelson Piquet	6.51 (1983–85)	+25.2	8	5–18
3	Nigel Mansell	6.45 (1986–88)	+22.3	9	5–21
4	**Alain Prost**	6.91 (1984–86)	+25.2	10	5–20
4	Jackie Stewart	5.51 (1970–72)	+10.7	10	6–22
4	Sebastian Vettel	7.16 (2011–13)	+17.2	15	6–37
4	Emerson Fittipaldi	7.40 (1974–76)	+17.2	16	7–34

Schumacher is the only driver alone in a tier, with a 70% chance of being first. He has both the best peak from 1970 and the biggest career total. Tier 5 runs from Andretti and Rosberg through Häkkinen, Button, Räikkönen and Norris. The peak figures here come from the 1970+ model, so they differ slightly from the all-time chart. Stewart is low because only his seasons from 1970 count.

Career, from 1970

Tier	Driver	Career total	Typical rank	90% range
1	**Michael Schumacher**	+56.5	2	1–4
1	Fernando Alonso	+52.5	2	1–6
1	Max Verstappen	+48.6	3	1–5
1	Lewis Hamilton	+44.4	4	1–8
2	**Ayrton Senna**	+36.6	5	3–7
3	**Niki Lauda**	+26.7	8	5–18
3	Nelson Piquet	+25.3	8	5–19
3	Alain Prost	+25.2	8	5–18
3	Nigel Mansell	+22.3	10	6–24

A career total of +50 is the same as finishing 5 places ahead of an average driver for 10 full 22-race seasons. Alonso's total comes from length as much as height: 445 starts, many of them in cars that couldn't win. Verstappen's total is still growing. Tier 4 includes Button, Vettel, Fittipaldi, Barrichello, Häkkinen and Rosberg.

The single strongest season

The strongest single season in the data is Michael Schumacher's 1994 with Benetton. In identical cars he would have averaged fourth place against every driver in F1 history. Both the all-time model and the 1970+ model rank it first. It is a narrow lead, though: it has only about a 9–13% chance of really being the best season. Fangio's 1953 and 1954 are about as likely.

Rank	Driver	Season	Team	Races	Season score	90% range	Chance it is the best
1	**Michael Schumacher**	1994	Benetton	14	4.03	2.7–5.4	9%
2	Juan Manuel Fangio ▲	1953	Maserati	8	4.16	2.6–6.0	11%
3	Juan Manuel Fangio ▲	1954	Mercedes and Maserati	8	4.26	2.7–6.3	11%
4	Jim Clark △	1965	Lotus	9	4.42	2.6–6.1	8%
5	Michael Schumacher	2000	Ferrari	17	4.44	3.0–6.2	6%

Rank	Driver	Season	Team	Races	Season score	90% range	Chance it is the best
6	Juan Manuel Fangio	1956	Ferrari	7	4.45	3.0–6.3	6%
7	Ayrton Senna	1988	McLaren	16	4.48	3.0–6.3	6%
8	Juan Manuel Fangio	1957	Maserati	7	4.52	2.8–6.3	7%
9	Jim Clark △	1967	Lotus	11	4.52	2.7–6.3	7%
10	Ayrton Senna	1987	Lotus	16	4.56	2.9–6.2	5%

The scores come from the all-time model (C50). "Chance it is the best" compares the 15 best seasons using their published ranges. A season counts if the driver started at least 8 races, or 60% of a short early calendar.

Why 1994 tops the list. Schumacher beat three Benetton teammates that year by a wide margin: Jos Verstappen, JJ Lehto and Johnny Herbert. The model reads that gap as his skill. There were claims at the time that the second Benetton wasn't equal and that the team used banned electronics. The FIA found banned software on the car but no proof it was used. If the second car really was weaker, some of the credit is the car's. Without 1994, the next best seasons from 1970 are Schumacher 1995 and Senna 1988, almost level.

The 1950s seasons carry a warning. Fangio's 1953 and 1954 are built on only 8 races each, and the 1954 Mercedes is one of the cars the model rates too low (section 8).

Most added to the car in one season. A different question is which driver lifted a real car furthest above what an average driver would have done with it. On that measure, from 1970, the leader is Niki Lauda in 1978. In the third-best car, the Brabham, he gained 5.6 places a race (90% range 3.1–8.4). Next come Lauda again in 1979, then Alonso's 2009 Renault, the eighth-best car of 10, at +4.3. Gains are larger in weaker cars, which have more places to climb, so this ranks seasons differently from the identical-car score.

6. Drivers without the best car

The model supports the idea that Alonso's 2012 was one of the best seasons in the data. Of all the leading drivers that year, he gained the most over an average driver in the same seat, in only the third-best car.

How "gain" is measured

For each season, the model keeps the real drivers in their real cars, then swaps one driver for an average driver in the same seat. The *gain* is how many places that driver was worth, on average, compared with the average driver. Gain has a ceiling: a car good enough for

second can gain at most one place. So gains in the very best cars are understated, and this measure is read alongside the identical-car scores rather than on its own.

2012 as a test case

The model was not tuned to produce any result for Alonso.

Driver	Car	Car rank in 2012	Places gained over an average driver	90% range
Fernando Alonso	Ferrari	3rd of 12	+3.54	+2.3 to +5.1
Lewis Hamilton	McLaren	2nd of 12	+3.11	+1.8 to +4.6
Sebastian Vettel	Red Bull	1st of 12	+1.92	+0.7 to +3.3
Jenson Button	McLaren	2nd of 12	+1.89	+0.6 to +3.2
Romain Grosjean	Lotus	4th of 12	+1.45	−0.5 to +3.3
Kimi Räikkönen	Lotus	4th of 12	+1.35	−0.4 to +3.1
Felipe Massa	Ferrari	3rd of 12	+0.13	−1.2 to +1.6
Mark Webber	Red Bull	1st of 12	−0.62	−1.7 to +0.7

Alonso's range overlaps Hamilton's and Vettel's, so this supports the view of 2012 without proving it. Much of the evidence is one teammate, Massa, whose own level is an estimate. Vettel's gain is held down by the ceiling, because he had the best car.

Best cars against lesser cars, career-long

Here the same drivers' seasons are split by car. A *lesser* car is one ranked 4th or lower that season, and only seasons with at least 8 starts count.

Driver	Seasons in a top-3 car	Average gain	Seasons in a lesser car	Average gain
Fernando Alonso	1	+2.96	16	+2.19
Max Verstappen	2	+1.54	3	+3.38
Ayrton Senna	3	+2.22	4	+3.11
Niki Lauda	5	+3.04	3	+4.01
Nelson Piquet	3	+2.46	8	+2.98
Nigel Mansell	4	+2.17	5	+2.38
Michael Schumacher	4	+1.78	5	+1.21
Lewis Hamilton	6	+1.35	2	+1.45
Alain Prost	6	+1.05	1	+0.93

Alonso stands out: 16 seasons in lesser cars, gaining over two places a season in them. No other leading driver comes close to that many. Gains are naturally larger in lesser cars, which have more room to climb, so this table is read across drivers rather than across its columns. Hamilton and Prost spent most of their careers in top cars, where the ceiling limits what this measure can show.

7. How far to trust it

The model passed four kinds of check: predicting races it never saw, recovering answers planted in fake data, matching the real constructors' championship, and settling to one answer when fitted. It is reliable on dry qualifying and race skill. It is much weaker on wet-weather and adaptability scores.

Predicting races it hasn't seen

Every fifth round of every season was hidden from the model during fitting. The model then predicted which of two teammates would finish ahead in those hidden races. It was compared with two simple rules: picking whoever had the better head-to-head record so far, and picking whoever started ahead.

Model	Session	Teammate pairs	Model correct	Head-to-head record so far	Started ahead
C50 (1950+)	Qualifying	1,962	70.7%	66.9%	–

Model	Session	Teammate pairs	Model correct	Head-to-head record so far	Started ahead
C50 (1950+)	Race	910	70.7%	59.2%	69.6%
B80 (1970+)	Qualifying	1,856	70.6%	69.5%	–
B80 (1970+)	Race	857	70.5%	54.5%	69.7%
B18 (2018+)	Race	250	74.8%	60.0%	72.0%
A18 (laps, 2018+)	Race	250	74.4%	60.0%	72.0%

The model beats both rules every time. For races it beats "started ahead" without being told the grid of the hidden race. About 70% is close to the ceiling for this question: teammates are usually close, and luck decides many of their battles.

Recovering known answers

Fake seasons were generated on the real calendar with driver skills I chose, then the model was fitted to them. Two things were checked: how well it recovered the true order (rank correlation, where 1.00 is perfect), and whether its 90% ranges contained the truth about 90% of the time.

What was measured	Order recovered (C50)	Order recovered (B80)	90% range hit (B80)
Qualifying skill	0.94	0.91	92%
Race skill	0.87	0.85	88%
Wet qualifying	0.42	0.30	87%
Wet race	0.41	0.42	82%
Changeable race	0.32	0.33	85%
Adapting to a new team	0.51	0.56	84%
Adapting to new rules	0.69	0.41	88%

The main skills come back well, so the rankings rest on solid ground. Wet, changeable and adaptability scores come back only weakly, so they show a direction but can't separate individual drivers. That is why none of them is a headline result. The ranges are about the right width, so the uncertainty shown is honest.

Matching the constructors' championship

The model never sees the constructors′ table, so it is an outside check on the car ratings. Each season, the model′s car order was compared with the real constructors′ order.

- The average rank correlation from 1970 is **0.88**.
- The car the model rates best finished first or second in the constructors′ in **96%** of seasons.
- The weakest stretch is 1970–75, at **0.61**, with 1971 the worst single season at 0.23. Fields then were full of one-off privateer entries.

Did the fit settle?

Yes. For the 1970+ model the worst R-hat on a main value is 1.013, against a usual limit of 1.05. The lap-time model was re-run with 25,000 steps per chain, which brought its worst R-hat to 1.0003. Longer runs remove computing noise but not the uncertainty in the data. Peak ranges stayed about four finishing places wide, which is a property of F1 history, not of the method.

8. Stress tests

The ranking survived every stress test without a full reshuffle. Each test below asks whether a plausible flaw is driving the results. All 48 tests, features and checks, with whether each went into the final model, are in the separate test register (f1-model-test-register.xlsx).

Can cars be dominant enough?

The worry. Models like this tend to pull car ratings towards the middle. If a truly dominant car (the 1988 McLaren, say) is rated too low, the leftover speed is credited to its drivers.

Test 1: planted dominance. In fake data, I added a known extra amount of speed to 21 historically dominant cars. The model gave 69–77% of it to the cars, and the rest leaked to the drivers. The drivers flattered most were Senna (by 0.11–0.21), Fangio (0.09–0.11) and Hamilton (0.05–0.12). Schumacher was flattered by only 0.02.

Tests 2–5: changing how cars are modelled. Each version was judged on hidden races (log loss, lower is better) and on the constructors′ check.

Change	Qualifying	Race	Constructors′ check	Verdict
Current model	.3584	.3684	.857	Kept

Change	Qualifying	Race	Constructors' check	Verdict
A few cars allowed to sit far from the rest (heavy tails, strength learned from the data)	.3584	.3686	.857	No gain
Very heavy tails, forced	.3588	.3690	not run	Rejected; fit didn't settle
Weaker year-to-year carry-over (.3 instead of about .8)	.3588	.3718	.816	Rejected, kept as an assumption test
Half the season-to-season change in driver form	.3592	.3684	.856	No gain

No version predicted better, so the main model is unchanged. The carry-over test still matters, because it moves the rankings more than anything else tried. With weaker carry-over, Senna becomes first on peak at 3.78, Hamilton improves from 6.59 to 4.62 and Lauda from 5.57 to 4.56. That version predicts worse and its fit didn't fully settle, so it isn't adopted. It is the main reason the tiers are drawn wide.

Does one teammate make or break a rating?

In 15 refits, I removed a leading driver's main teammate from the races they shared, refitted the whole model, and measured how far the driver moved. The Senna–Prost refit appears twice, once for each driver. Two fits with nothing changed differ by about 0.11, so moves that size are noise.

Driver	Teammate removed	Shared races	Peak change	Peak rank
Nigel Mansell	Riccardo Patrese	46	−0.83 (better)	13th → 9th
Alain Prost	Ayrton Senna	32	+0.78 (worse)	18th → 28th
Lewis Hamilton	Nico Rosberg	78	−0.64 (better)	14th → 9th
Graham Hill	Richie Ginther	29	−0.48 (better)	–
Michael Schumacher	Rubens Barrichello	104	+0.21	2nd → 4th
Fernando Alonso	Lance Stroll	89	−0.21	–
Fernando Alonso	Felipe Massa	77	+0.17	10th → 13th

Driver	Teammate removed	Shared races	Peak change	Peak rank
Jim Clark	Trevor Taylor	19	+0.16	–
Juan Manuel Fangio	Karl Kling	11	+0.15	stays 1st
Max Verstappen	Sergio Pérez	90	+0.13	–
Sebastian Vettel	Mark Webber	94	+0.11	–
Ayrton Senna	Gerhard Berger	48	+0.11	–
Lewis Hamilton	Valtteri Bottas	100	−0.11	–
Ayrton Senna	Alain Prost	32	−0.10	–
Jackie Stewart	François Cevert	46	+0.04	–
Niki Lauda	Clay Regazzoni	57	+0.03	–

Most leading drivers don't depend on one teammate: in 12 of the 15 refits the leading driver moved by 0.21 or less. Two patterns stand out.

- **Prost needed Senna.** His two seasons nearly matching Senna in 1988–89 were the strongest evidence that he was elite. Without them his peak drops by 0.78, even though his best seasons, 1984–86, weren't touched.
- **A driver alone in a strong car gets flattered.** Hamilton without Rosberg, Mansell without Patrese and Hill without Ginther all *improve*. Once the teammate is gone, nobody else drove the 2014–16 Mercedes, 1991–92 Williams or 1962–64 BRM closely enough to compare. The model then can't split car from driver directly, so it credits part of the car to the driver. So drivers who had a close teammate in their best car are, if anything, rated a little harshly next to those who didn't.

Does it agree with lap times?

The main model sees only positions. From 2018 there are lap times it never used, so a separate model built on laps is an independent check.

Check, 2018–2026	Agreement
Car rating against the team's dry qualifying pace (91 team-seasons)	.95
Car rating against the team's dry race pace	.93
Teammate gap against the qualifying lap-time gap (94 pairings)	.79; same driver ahead 77% of the time

Check, 2018–2026	Agreement
Teammate gap against the race lap-time gap (93 pairings)	.82; same driver ahead 80% of the time
Season scores against the lap-time model (183 driver-seasons)	.90
Peak ranking from 2018 against the lap-time model (25 drivers)	.91

Both models put Verstappen first and Hamilton second from 2018, with Leclerc and Norris next. The biggest disagreement is at the back. The lap model rates Russell and Albon at Williams, Mick Schumacher at Haas and Gasly higher, because finishing 18th instead of 19th says little about how well someone drove. So the results model is most reliable for drivers who spent their best years in competitive cars. That covers almost everyone near the top.

Were early eras easier to dominate?

The 1950s had huge gaps between the front and the back of the grid. If the gap between drivers was also bigger, early stars might look better than they were. Two things argue against a big effect.

1. Every driver is scored in one shared field across all eras, so weak 1950s privateers don't flatter Fangio.
2. Once cars are removed, drivers were only about 10% more spread out in the 1950s and 60s than later (a standard deviation of 2.30–2.35 against 1.99–2.17).

Shrinking each decade's spread to the all-time average left tier 1 unchanged. Only its order shuffled: Senna 1st, Schumacher 2nd, Stewart 3rd, Fangio 4th and Clark 5th. Across all drivers the order barely changed (correlation .999). No adjustment was made.

Is a car's speed being credited to its early driver?

This is the biggest weakness in the all-time ranking. Before the late 1960s, fields were small, teams often had one or two regular drivers, and many entries were one-offs. When the model rates a car lower than it really was, the missing speed goes to the driver. For each early driver near the top, the model's car rank in his peak seasons was compared with where the team really finished on points.

Driver	Model's car rank / team's real rank	Warning	Why
Juan Manuel Fangio	1954 Mercedes 3rd / 1st	▲ Likely inflated	The car won 4 of the 6 races it entered. Fangio was also flattered in the planted-dominance test.
Graham Hill	1964 BRM 6th / 1st; 1965 BRM 4th / 1st	▲ Likely inflated	Removing Ginther improved Hill by 0.48, which fits a strong car being credited to him.
Jim Clark	1966 Lotus 7th / 5th; 1967 3rd / 2nd	△ Possibly inflated	In the planted-dominance test Clark moved the other way, so the evidence is mixed.
Alberto Ascari	1951–53 Ferrari 1st / 1st	△ Possibly inflated	Ferrari won 14 of 15 races in 1952–53, and the model only gives a car about 70–75% of a real dominance.
Nino Farina	1950 Alfa Romeo 1st / 1st	△ Possibly inflated	The Alfa won all 6 races in 1950. It's the same leak as Ascari's.
Jack Brabham	1966 Brabham 2nd / 1st	△ Possibly inflated	Rated second-best, but won the championship.
Rindt, Surtees, Moss	At or above their real results	Not flagged	Their cars are rated at least as well as they performed.

A team's points include its driver's own results, so a great driver lifts his team's finishing place. Some of each gap really is the driver, and the model can't settle how much. These are warnings, not corrections, and the scores are unchanged.

Changing the assumptions

Each assumption was changed in turn and the model refitted. The order was compared with the main run, where 1.00 means identical.

Assumption changed (1970+ model)	Peak order	Career order
Team orders not corrected	.99	.99
Adapting lasts about 12 races, not 6	.99	.98
Collisions counted as driver errors	.99	.98

Assumption changed (1970+ model)	Peak order	Career order
Customer teams not separated from works teams	.99	.99
No warm-up seasons before 1970	.98	.98
Wet and changeable merged	.99	.99
Qualifying time gaps added (1950+ model)	.991	.976
Weaker car carry-over (1950+ model)	.96	.89

The overall order hardly moves, but the very top is sensitive. Three of these changes put Senna first on peak, counting collisions as errors puts Alonso first on career, and adding qualifying times drops Schumacher from 2nd to 5th on all-time peak. This is why the result is given as tiers.

9. Limitations

The ranking is only as good as its weakest links, and these are the ones to keep in mind when reading it.

- **Comparing eras rests on chains of teammates.** A 1950s driver connects to a 2020s driver only through decades of overlapping careers. The gap between distant eras is the least certain number in the project.
- **Positions can't show margins.** The model knows who finished ahead, not by how much. A driver who won by a lap and one who won by a tenth get the same credit. Adding qualifying time gaps and race margins was tested. The results were mixed, so both are left out.
- **Back-of-the-grid drivers are measured poorly.** Finishing 18th rather than 19th says little. The lap-time check confirms that drivers in slow cars are where finishing positions say least.
- **A lone driver in a strong car is flattered.** Without a close teammate the model can't split car from driver, as the teammate-removal tests showed for Hamilton, Mansell and Hill.
- **Early data is thin.** Before the late 1960s, small fields and one-off entries make early car ratings less reliable. Fangio and Graham Hill are probably flattered.
- **Wet and adaptability scores are rough.** They recover only weakly in known-answer tests and are not used to separate individual drivers.
- **Some inputs are hand-built.** Pre-2018 wet shares are estimated from race reports. The 16 team-order cases come from known incidents and weren't checked race by race.

- **Reliability isn't rewarded.** Mechanical failures are removed, so a driver who nursed a fragile car home gets no credit for it.
- **Careers before 1970 are cut off.** Career and combined rankings start in 1970, so Stewart, Ickx and Regazzoni are scored only on their later seasons. The all-time peak ranking doesn't have this problem.
- **Career totals reward longevity.** That is deliberate, which is why they are reported next to peak rather than instead of it.

10. Conclusion

Once the car is taken out, Michael Schumacher has the strongest overall claim to be the greatest Formula 1 driver. He is the only driver in the top tier on all three measures (peak, career and combined), and the only one alone in his tier, with a 70% chance of ranking first on the combined measure. His claim held up when a teammate was removed and when qualifying times were added. In both cases his peak slipped a few places, but he stayed in the top tier.

At their best, five drivers can't be separated: Fangio, Senna, Clark, Schumacher and Stewart. Among them, Senna has the cleanest case. His rating survives every stress test, and several reasonable changes to the model put him first. Fangio and Clark sit at the very top of the estimates, but some of their credit probably belongs to cars the model rates too low. That is especially likely for Fangio's 1954 Mercedes.

The model backs the case for drivers who lacked the best car. Alonso gained more over an average driver than anyone else in 2012, in the third-best car. He spent 16 seasons in lesser cars, gaining more than two places a season in them, and sits in the top career tier with Schumacher, Verstappen and Hamilton. Verstappen is the strongest driver of the current era, already level with that group on career and still adding to it.

The main lesson is about how precise any answer can be. Seventy-six years of results can separate tiers of drivers but not individual places at the top. The model rules out some popular claims: for example, that a driver who won titles in the best car must be better than one who didn't. It can't settle a debate between Senna and Schumacher, and it shouldn't be used to. A fair answer to "who was the greatest?" names a small group, gives the evidence for each, and says where the evidence runs out.

Appendix

Glossary

Term	Meaning
Peak score	Average finishing position in identical cars, in a 20-car field, over a driver's best three consecutive seasons. Lower is better; an average driver scores about 10.5.
Career score	Sum over seasons of (10.5 − season score), weighted by races started, with 22 races counted as one season.
Gain	Places a driver was worth over an average driver in the same car in a real season.
90% range	The span a value or rank falls in nine times out of ten, given the data.
Tier	A group of drivers who can't be reliably separated: each has at least a 25% chance of being better than the tier's leader.
Plackett–Luce	A model for ranked results that picks the winner, then second place, and so on, each in proportion to performance.
Log loss	How surprised the model is by results it didn't see; lower is better.
R-hat	Checks that independent runs of the fit agree; 1.00 is perfect, below about 1.05 is acceptable.
Rank correlation	How closely two orderings agree; 1.00 means identical.

Files

- **f1-model-test-register.xlsx**: every test, feature and check (48 rows), marked in the model, not in the model, assumption test or check only, with the teammate-removal results on their own sheet.
- **f1-model-walkthrough**: the interactive results page with every season score and table.
- **f1model-v2-final.zip**: the code, inputs and result tables.

Sources

- F1DB: official results, 1950–2026.
- TracingInsights: lap times, 2018–2026.